

PAPER_1708 — ITER Edge Safety Factor $q_{\text{edge}} = q_{\text{edge}} = K_{\text{MEX}} - F \cdot \Phi = 25/12 - 1/12 = 2$ (avoids m/n=2/1 kink)

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Plasma Fusion **Date:** June 18, 2026 **Status:** CLOSED — EXACT closure (PAPER_1196 plasma fusion proof set)

Observation

PAPER_1196_S418 (PAPER_1196 Plasma Fusion Unified Proof Set) gives:

$$q_{\text{edge}} = K_{\text{MEX}} - F \cdot \Phi = 25/12 - 1/12 = 2 \text{ (avoids m/n=2/1 kink)}$$

Status: **EXACT**.

UQFF Closed Identity

$$q_{\text{edge}} = K_{\text{MEX}} - F \cdot \Phi = 25/12 - 1/12 = 2 \text{ (avoids m/n=2/1 kink)} \quad \text{EXACT}$$

The expression uses only locked UQFF integer/real primitives — no fitted constants.

NOT REPLACEMENT

ITER design parameters are chosen by international fusion-engineering consensus. UQFF supplies structural integer-primitive forms demonstrating the parameters' deep mathematical origin.

Reference

- Source: PAPER_1196_S418
- Related: PAPER_1696-1705 (tier-28); PAPER_1686-1695 (tier-27)
- Calculator dispatch: `calculate_paradox({"paradox": "q_edge_2"})`

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